

CAMILLE TRAN

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Career Objective

Data Science graduating senior (June 2026) with practical experience spanning across data engineering, machine learning, and data visualization gained through industry-focused projects and internships. My experience includes building end-to-end ML pipelines, real-time streaming systems, and multi-agent AI applications. I'm seeking opportunities where I can apply my strong MLOps foundation and multi-cloud experience. I bring a tactical approach, positive "can-do" attitude, and great communication skills.

Education

University of California - San Diego, Major GPA: 3.64

La Jolla, CA

Bachelor of Science in Data Science - Hahcioğlu Data Science Institute

Expected Graduation: Spring 2026

Senior Project 🌀 (Fall 2025 – Winter 2026): Building a stress detection app using ML to distinguish cognitive stress from physical activity by analyzing physiological biomarkers, providing personalized health insights from smartwatch data.

Technical Skills

Languages & Libraries: Python (pandas, NumPy, Matplotlib, Seaborn), R, Java, C++, SQL, Scala, HTML/CSS, MATLAB

Tools & Frameworks: TensorFlow, PyTorch, Scikit-Learn, Keras, Apache Spark (PySpark, Spark), SQL, Apache Airflow, Kafka, Hadoop (HDFS, Hive), NiFi, FAISS, Kubernetes, Docker, Git, Linux, Jupyter, Streamlit, Grafana, Tableau, Power BI

Cloud Platforms: AWS (SageMaker, Glue, Athena, CodePipeline, ECR, S3, EC2), Snowflake, Azure (Logic Apps, SQL DB)

Certifications: AWS Cloud Practitioner (2023), AWS Data Engineer–Associate (2025), AWS Machine Learning (In Progress)

Experience

US State Department - US Embassy London

September 2023 – May 2024

AI/Data Science Developer Intern

- Designed an AI-driven document retrieval pipeline for policy Q&A, integrating LangChain for text preprocessing and FAISS for fast vector search across government documents.
- Optimized embedding-based retrieval to improve query accuracy and reduce latency, enhancing search responsiveness.
- Participated actively in Scrum teams, during weekly and daily meetings collaborating effectively to meet project goals.

Projects (click 🌀 to view project)

Multi-Agent AI Assistant System 🌀 | *AutoGen, GPT-4o, Tavily API, SQLite*

- Built a multi-agent AI system with AutoGen 0.6.1 and GPT-4o featuring a five-layer architecture (UI, Memory, Orchestration, Agents, External Services) with 2 specialized agents delivering sub-5-second response times.
- Developed an interactive CLI, SQLite-based conversation memory with session tracking, and structured logging.
- Integrated three external APIs (OpenAI, Tavily Search, Open-Meteo) with error handling and secure key management.

Gaussian Time Series MLOps Pipeline 🌀 | *AWS SageMaker, Terraform, CodePipeline, Flask, ECR*

- Developed end-to-end MLOps pipeline on AWS using Gaussian Process Regression that achieved $R^2 = 1.0$ and MAE of 10.4 across 132 months of historical call center data, with fully automated CI/CD through CodePipeline.
- Implemented infrastructure-as-code with Terraform for reproducible ECR/Lightsail deployments and Flask API.
- Staged deployment with manual approval gates between staging and production environments.

Clinical Analytics with ML Drift Detection 🌀 | *Snowflake, Scikit-Learn, Power BI*

- Developed Snowflake healthcare analytics processing 236K+ patient records with ML drift detection using 5 statistical tests (K-S, PSI, Chi-Square) and automated retraining at 10% performance degradation.
- Created Power BI dashboards for real-time model monitoring and clinical KPIs with automated retraining.
- Built alerting system for managing data quality and feature distribution shifts in production environments.

Real-Time Big Data Pipeline on AWS 🌀 | *Kafka, PySpark, Hive, Apache Airflow, NiFi*

- Programmed a real-time streaming pipeline with Kafka, PySpark Structured Streaming, and Hive for analytics storage.
- Orchestrated fault-tolerant workflows using Apache Airflow DAGs with automated task scheduling, managing 15+ dockerized services (Kafka, Spark, Hive, HDFS) for real-time data warehousing and trend analysis.
- Implemented data lineage with Apache NiFi for secure ingestion and encryption across distributed pipeline.

Real-Time E-Commerce Observability Stack 🌀 | *Kafka, Spark, InfluxDB, Grafana*

- Built observability stack with Confluent Kafka (Avro), Spark Streaming, and InfluxDB for e-commerce metrics.
- Created Grafana dashboards with real-time alerts for revenue tracking and system health monitoring.
- Implemented a Docker-based stack with Docker Compose and Schema Registry for reproducible deployments.